

9(a)	1 individuals (within a population) can, breed / mate / reproduce ; 2 to produce fertile offspring ; 3 are reproductively isolated (from other species) / AW ;	3
9(b)	<i>any one from:</i> 1 for organisms that, do not breed sexually / are asexual ; 2 for, fossil / extinct, organisms ; 3 morphological differences are easier to determine ; 4 often not possible to observe reproductive behaviour ; 5 time required for mating behaviour may take too long (to be observed) / not possible to achieve conditions in which mating can occur (while being observed) ;	1

Question	Answer	Marks
9(c)	any four from: <i>max three:</i> 1 both are prokaryotes or - both have no, nucleus / (double) membrane-bound organelles ; 2 both have circular DNA ; 3 both have 70S ribosomes ; 4 both reproduce by, binary fission / budding / fragmentation ; <i>max three:</i> 5 bacteria cell wall made of peptidoglycan whereas Archaea cell wall does not have peptidoglycan ; 6 bacteria cell membrane has ester-linked lipids whereas Archaea cell membrane has ether-linked lipids ; 7 have different ribosomal RNA ; 8 Bacteria do not have (DNA associated with) histones whereas Archaea do have (DNA associated with) histones ; 9 Bacteria form spores whereas Archaea do not form spores ;	4